MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, OIL, SYNCHO, 0-200 PSI 320-DEGREE MOVEMENT, 26V, SINGLE PHASE, 400 CYCLES

This specification is approved for use with the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating 320-degree movement, synchro style, differential, oil pressure transmitter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-T-5350	Transmitters, Synchro Operated, Aircraft, General Specification For
MIL-I-7087	Indicator, Pressure, Oil, 0-200 PSI

STANDARDS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620
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MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-2073-1	DOD Material Procedure For Development And Application Of Packaging Requirements
MS28005	Transmitters, Pressure, Synchro Aircraft
MS28010	Indicators, Pressure-Synchro, Single, 2-Inch Size

(Copies of specification, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadephia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

3. REQUIREMENTS

3.1 Qualification. Item furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.2).

3.2 General. The requirements specified in Specification MIL-T-5350 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2.1 Precedence. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.3 Design. The oil pressure transmitter shall be design for use with an MS28010-3 pressure indicator conforming to Specification MIL-I-7087.

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3.3.1 Hermetically sealed case. The case shall provide a Hermetically sealed enclosure for the entire internal mechanism of the transmitter. The design of the case shall permit removal and replacement of the internal mechanism and resealing of the case. This shall be accomplished without the use of special tools and fixtures unless they are approved by the contracting activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the transmitter may be subjected.

3.4 Construction.

3.4.1 Dimensions. The outline dimensions of the transmitter shall conform to Standard MS28005.

3.4.2 Overpressure stop. An overpressure stop shall be provided and shall be designed to restrain the prime mover. The stop shall allow the pointer of any indicator to travel beyond full scale to a point approximately midway between the last and the zero graduation.

3.4.3 Bosses. The bosses, as shown on Standard MS28005, shall be provided for external pressure and vent connections to the pressure sensitive element and the opposing pressure element, respectively. The bosses shall be identified with white lettering "P" and "V" permanently inscribed 0.125 inch in height.

3.4.4 Restrictors. No restrictors or capillary line connecting the pressure fitting to the pressure element shall be used which will introduce an objectionable lag in the indication response.

3.5 Weight. The weight of the transmitter shall not exceed 1.4 pounds.

3.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130. The pressure range of the transmitter shall be included as Special Characteristics as specified in MIL-STD-130.

3.7 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program.

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The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.5).

b. Quality conformance inspection (see 4.6).

4.3 Test conditions. Unless otherwise specified, the transmitter shall be tested with the longitudinal axis in a horizontal plane.

4.4 Tolerances. When the transmitter is subjected to the tests specified in Specification MIL-T-5350, the tolerances shall not exceed those specified in tables I and II.

4.4.1 Scale error. When testing for scale error in accordance with Specification MIL-T-5350, the transmitter shall be tested first with increasing readings and then with decreasing readings. With the readings increasing, the indication shall be brought up to, but not exceeding, the indication specified to give the desired readings. With readings decreasing, the indications shall be brought down to, but not fall below, the indication specified to give the desired reading.

TABLE I
Tolerances (psi)

Test (psi)	Room temperature	Vibration, overpressure	Low temperature, high temperature, seasoning endurance, and altitude tests
0	3.0	4.0	5.0
50	3.0	4.0	5.0
100	4.0	5.0	6.0
150	4.0	5.0	6.0
200	4.0	5.0	6.0

TABLE II
Oil pressure transmitter

Test	Tolerance-(psi)
Friction:	
Room temperature	3.0
High temperature	5.0
Cold temperature	5.0
Position error at 100 psi	1.0
Vibration	
Pointer oscillation	3.0

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4.5 Qualification inspection. Three qualification inspection samples representative of production item shall be inspected in accordance to paragraph 4.5 of MIL-T-5350 and as specified herein.

4.6 Quality conformance inspection. Quality conformance inspection shall be accomplished as called out in MIL-T-5350.

4.7 Special tests.

4.7.1 Overpressure. The transmitter shall be subjected to a pressure of 400 pounds per square inch (psi) for a period of 10 minutes. The transmitter shall be tested not less than 1 hour following the overpressure test for scale error as specified under individual tests on increasing pressure only. The scale error shall not exceed the tolerance specified in table I.

4.7.2 Damping. A pressure sufficient to produce full scale pointer deflection shall be applied to the transmitter. A quick release valve shall be used to exhaust the pressure. The time required for the pointer to traverse the interval from 200 psi to 20 psi shall be within the period of 9 $\pm$ 3 seconds. The test shall be conducted with lubricating oil with a viscosity of approximately 2,500 centistokes, such as grade 1005 oil at -65°F, or equivalent.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each Transmitter shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each transmitter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

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5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The oil pressure transmitter covered by this specification is intended to transmit the oil pressure of the lubricating system of aircraft engines to an applicable indicator.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Applicable level of packaging and packing (see 5.1).

(c) With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the Qualified Products List QPL-5790 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/MMEDO TAFB OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity."

6.3 Subject term (key word) listing.

Oil
Pressure
Synchro
Transmitter
10G
26 VAC
0-200 PSI

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6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing activity:
AIR FORCE -71

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-5790E(USAF)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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